Ji et al. 2020 — Cutaneous Squamous Cell Carcinoma

Cutaneous SCC · 10x Genomics · 10,598 malignant cells (8/10 samples ≥100 → 10,531; developmental compartment 6,897 after scoping the cycling lobe).

Question: do malignant cells recapitulate embryonic programs (Gast / pSom / Crest), and does the Gast/pSom hub→two-arm bifurcation reproduce?

GO / NO-GO GATES

Gate 1 (integration + continuity) — **GO**

Harmony(theta=5) on sample over 20 PCs: patients thoroughly intermixed, one connected kNN component, one continuous body. A cycling lobe (35% of cells) is a proliferation confound → scoped to the developmental compartment for the trajectory.

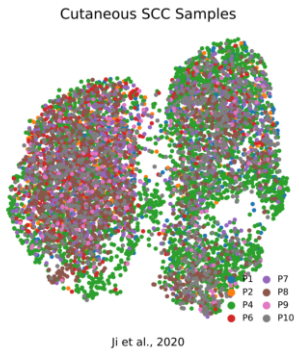
Gate 2 (embryonic module pattern) — **GO — non-branching**

Modules present and structured, but NON-branching: Gast & pSom OVERLAP (smoothed r=+0.11) and JOINTLY oppose Crest ((Gast+pSom)–Crest r=−0.88). No Gast-arm vs pSom-arm divergence — a principal-tree Y has both terminal arms leaning the same way (Gast<pSom); Crest = inverse terminal gradient. Matches the linear Sun/Kürten references, not the branching Puram hub-and-spoke.

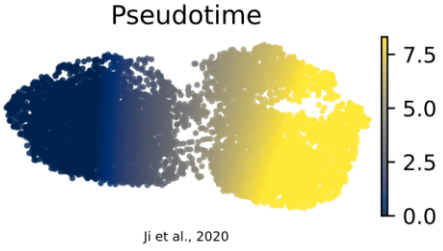
Gate 3 (cross-repression, branching only) — **NA**

Not applicable — no bifurcation. (Context: module cross-repression on the Mixed convergence zone is r=−0.50, near the Puram anchor ≈−0.49 — the §1 "some cross-repression may persist" clause within the overlap case, not sufficient for branching without two spatially-resolved arms.)

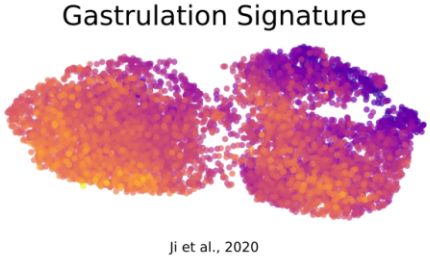
Samples (Gate 1: integration)



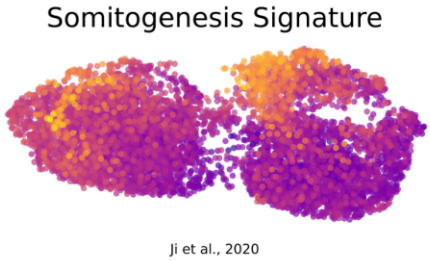
Manifold — pseudotime



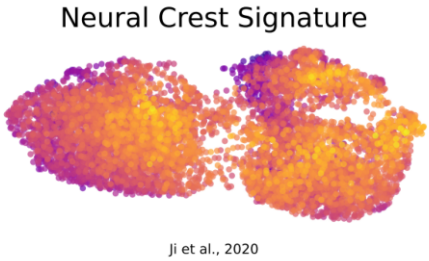
Gastrulation



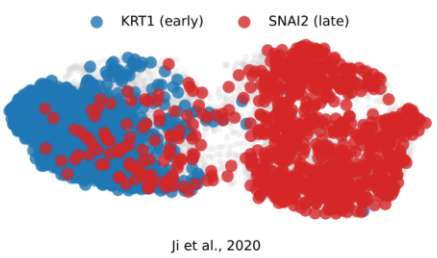
Somitogenesis



Neural Crest



Lineage-TF expression



KEY PARAMETERS & VERDICT

Min-cell threshold ≥100 (clean 6.7× count gap 268→40; keeps 8/10). Integration Harmony on sample, theta=5 (swept 0/2/5/10, visual mixing); PCs=20, no scaling. Cycling lobe scoped by per-Leiden cycling fraction >0.5 (3,634 cells; Gast-independent). Geometry NON-BRANCHING — principal curve (ElPiGraph), 2 tips / 0 forks, fold-free (ratio 2.97). Root Mixed (Gast/pSom-high, Crest-low origin — a tip, not the Crest-high terminus). Pseudotime gradient Gast ρ=−0.50, pSom ρ=−0.39, Crest ρ=+0.31. Annotation: data markers (KRT1/10, MMP10/PTHLH/LAMC2, TP63) + Ji 2020 subtypes (TSK, Tumor-KC Basal/Diff/Cyc). EMT-TFs shown: KRT1 (early) vs SNAI2/SLUG (late).

VERDICT: Embryonic programs reproduce as a NON-BRANCHING gastrulation/somitogenesis → neural-crest gradient (Gate 1 GO · Gate 2 GO non-branching · Gate 3 NA) — not the Puram-type bifurcation.